

Experiment 1: Installation of VirtualBox and Creation of a Virtual Machine

Aim

To install Oracle VM VirtualBox and create an Ubuntu Linux virtual machine that provides an isolated environment for cloud-computing laboratory experiments.

Tools Required

- Windows host computer
- Oracle VM VirtualBox
- Ubuntu Desktop LTS ISO image
- Minimum 4 GB RAM for the Ubuntu VM
- Minimum 25 GB dynamically allocated virtual disk

Detailed Procedure

Download and install Oracle VM VirtualBox on the Windows host using the Windows installer and default options.

Download an Ubuntu Desktop LTS ISO image.

Open VirtualBox Manager and select New.

Set the VM name to ubuntu-lab (or the chosen lab VM name), Type to Linux, and Version to Ubuntu (64-bit). Select the downloaded Ubuntu ISO.

Allocate 4096 MB (4 GB) RAM and create a VDI virtual hard disk with dynamically allocated storage of 25 GB.

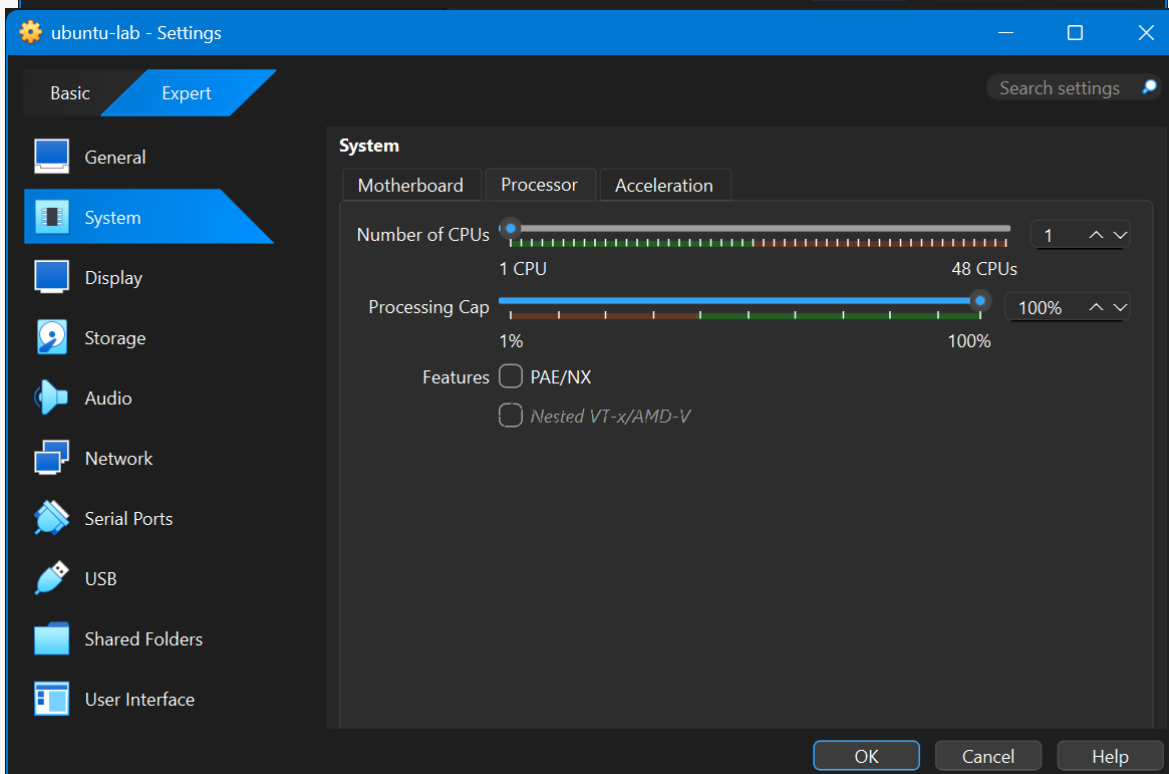
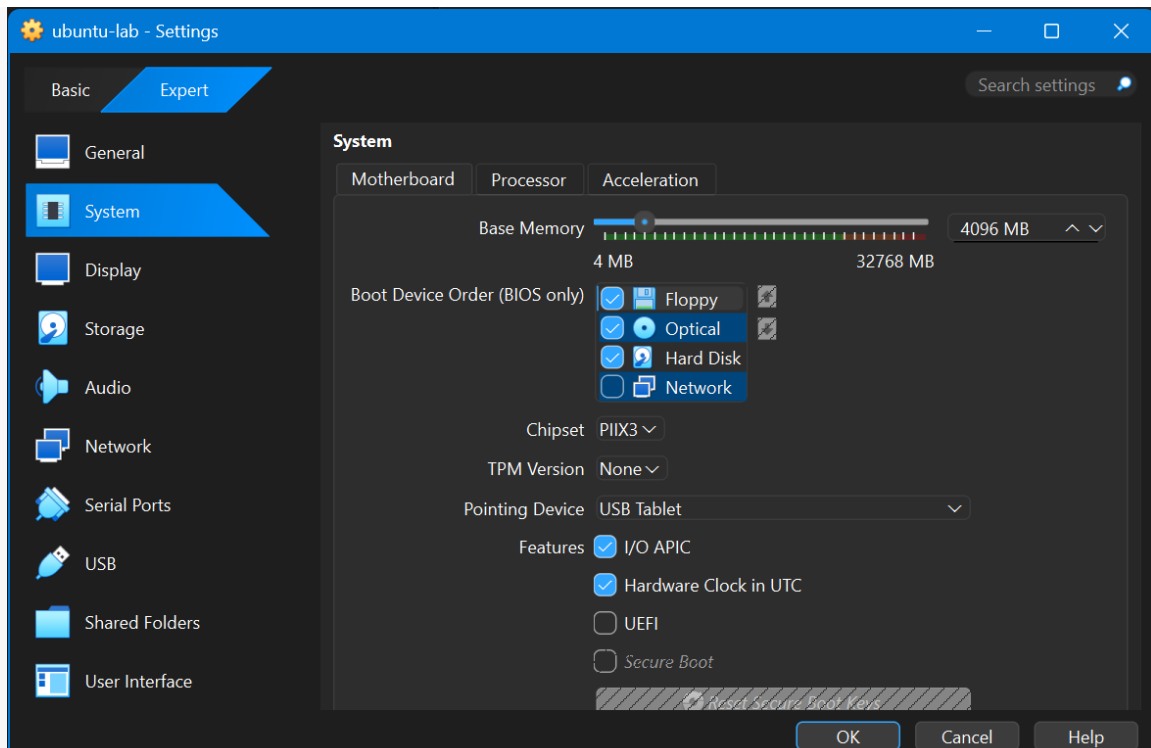
Finish VM creation. Open Settings → Storage and verify that the Ubuntu ISO is attached to the optical drive.

Start the VM and complete the Ubuntu installation. Select Install Ubuntu, choose the keyboard layout, and allow installation to complete.

Restart the VM when prompted and verify that Ubuntu starts successfully.

Output





Result

Ubuntu was successfully installed and booted inside VirtualBox, creating the isolated Linux environment required for subsequent experiments.